Cross-scale excitability in neuronal systems

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Abstract

An *in silico* network of quadratic integrate-and-fire neurons is built up from the single-cell level, and studied in the case of constant external inputs. Excitability is quantified by computing the spiking frequency at steady-state versus the injected DC current (the "f-I curves"). A mean-field model for globally-coupled neuronal populations is investigated through numerical simulations and mathematical analyses. In the case where both excitation and inhibition is present, the effect of network balance on excitability is quantified through a generalized f-I curve. This work shows that biologically-motivated network structures yield non-trivial excitability relationships which share common properties with those of the single-cell.

Introduction

The primary information-processing unit in the nervous system is the neuron. There are billions of neurons in the brains of mammals. These electrically-excitable cells consist of a cell body to process information, an axon to send messages, and dendrites to receive messages; they can have quite complicated structures (see figure 1).

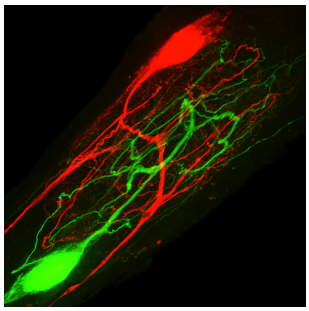

Figure 1: Confocal microscope image of two stained neurons (one red and one green) in the stomatogastric ganglion of the crab *Cancer borealis*, taken by Dr. Wolfgang Stein.

Neurons exchange information via all-or-none signals known as "action potentials," which consist of brief (a few ms) $\sim$100 mV depolarizations in the membrane voltage. Action potentials propagate along the length of the neurons to eventually communicate with muscles, glands, or other neurons. A realistic model of action potential generation was famously published by Alan Hodgkin and Andrew Huxley in 1952 [6]. This detailed model consists of 4 ordinary differential equations that describe how the conductances of ion channels in the neuronal membrane amplify inputs of sufficient quality through a nonlinear dependence on the voltage. Although incredible, the Hodgkin-Huxley (HH) model becomes computationally expensive to simulate as the number of neurons increases. Given its dimensionality, it is also difficult to understand the HH model with the powerful mathematical techniques that have been developed in 1, 2, and 3 dimensions. For these reasons, researchers have created more tractable mathematical models of action potentials, capturing the essence of the Hodgkin-Huxley model in lower-dimensional state spaces [3, 9].

The low-dimensional analog to the HH model studied here is the quadratic integrate-and-fire (QIF) neuron, which Hoppensteadt and Izhikevich have shown to be canonical [7]. In this context, "canonical" means that biophysically-accurate models like the Hodgkin-Huxley neuron can be transformed into the QIF neuron through a suitable change of variables. As is shown in the methods section of the present work, it is straightforward to understand how the QIF model describes excitability at the scale of a single cell. More challenging, however, is to understand the excitability of large groups of neurons. Recently, Avitabile *et al.* have made progress on this question by studying large, globally-coupled networks of neurons [1]. Their work focused on understanding the excitability properties of a single excitatory network, by analyzing special solutions called "canards." Informed by the related work of

Montbrió *et al.* [11], the goal of this article is to build up to an assessment of the excitability of a communicating pair of globally-coupled neuronal populations, one excitatory and one inhibitory. This is done by computing the relationship between the asymptotic frequency and injected DC current for a given neuronal system. Such a practice was pioneered by Hodgkin over 70 years ago [5] and comprises a foundation on which time-dependent stimuli can be understood.

Methods

All numerical simulations performed here have been with the forward Euler method, on an Acer PC with an AMD Ryzen 3 3200U 2.6 GHz processor. Computer codes for producing the various figures are available upon request, just email the author at zmobille3@gatech.edu.

Quadratic integrate-and-fire (QIF) neuron

One way of quantifying a neuron's excitability is by investigating its asymptotic firing frequency as a function of the injected DC current [8]. Such a relationship is known as the f-I curve. The purpose here is to work towards a generalization of this notion of excitability from the single neuron to networks of neurons. Since a network of neurons is composed of numerous single neurons, it will first be necessary to understand the excitability of a single neuron. To address this, let us consider the following model neuron:

$$\dot{V} = I + V^2, \quad \text{if } V \geq V_{\text{peak}} \text{ then } V \to V_{\text{reset}} \tag{1}$$

where V is the membrane voltage of the neuron, I is a constant parameter describing the injected DC current, and V_{peak} is the peak value of voltage which induces an instantaneous reset of the voltage to V_{reset}. Let us always assume that $V_{\text{peak}} > V_{\text{reset}}$. This model is known as the quadratic integrate-and-fire (QIF) neuron. Although simple, it captures much of the essence of the excitability properties of neurons. Let us calculate the f-I curve for the QIF neuron. The period of the QIF neuron, denoted as T, is the inverse of the frequency f and is the amount of time it takes for the voltage V to evolve from V_{reset} to V_{peak}. From equation 1, this statement is represented symbolically as

$$\frac{\mathrm{d}V}{\mathrm{d}t} = I + V^2$$

$$\frac{\mathrm{d}V}{I + V^2} = \mathrm{d}t$$

$$\int_{V_{\text{reset}}}^{V_{\text{peak}}} \frac{\mathrm{d}V}{I + V^2} = \int_0^T \mathrm{d}t$$

Noting that $\int (1 + x^2)^{-1} \mathrm{d}x = \arctan x$, make the change of variable $V = \sqrt{I}\tan\theta$. The integral for the period above then becomes

$$T = \int_{\theta_{\text{reset}}}^{\theta_{\text{peak}}} \frac{\sqrt{I}\sec^2\theta \mathrm{d}\theta}{(\sqrt{I}\tan\theta)^2 + I}$$

here $\theta_{\text{reset}} = \arctan(V_{\text{reset}}/\sqrt{I})$ and $\theta_{\text{peak}} = \arctan(V_{\text{peak}}/\sqrt{I})$. Using the identity $\sec^2\theta \equiv \tan^2\theta + 1$ and simplifying, this becomes

$$T = \frac{1}{\sqrt{I}} \int_{\theta_{\text{reset}}}^{\theta_{\text{peak}}} \mathrm{d}\theta$$

$$T = \frac{1}{\sqrt{I}} \left[\arctan\left(\frac{V_{\text{peak}}}{\sqrt{I}}\right) - \arctan\left(\frac{V_{\text{reset}}}{\sqrt{I}}\right) \right]$$

which indicates that the f-I curve of the QIF curve is given by

$$f = \sqrt{I} \left[\arctan\left(\frac{V_{\text{peak}}}{\sqrt{I}}\right) - \arctan\left(\frac{V_{\text{reset}}}{\sqrt{I}}\right) \right]^{-1} \tag{2}$$

Note that the frequency is defined only for $I > 0$. This is because the QIF neuron does not spike when $I < 0$; the frequency is zero for negative injected currents. In the limit that $V_{\text{reset}} \to -\infty$ and $V_{\text{peak}} \to \infty$, equation 2 above becomes

$$f = \frac{\sqrt{I}}{\pi} \tag{3}$$

The f-I curves given by equations 2 and 3 are compared against one another in figure 2. It is clear that the QIF neuron with finite V_{reset} and V_{peak} has a greater frequency than the neuron with infinite V_{reset} and V_{peak}, for all positive injected DC currents. The expression $\sqrt{I}/\pi$ therefore gives a lower bound on the frequency of the standard QIF neuron given by equation 1.

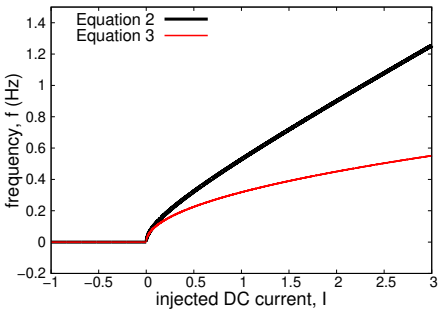

Figure 2: Equations 2 and 3 with $V_{\text{reset}} = -1$, $V_{\text{peak}} = 2$, showing class I excitability in the standard QIF neuron ($V_{\text{reset}} < 0$).

A qualitatively different f-I curve is obtained by setting $V_{\text{reset}} > 0$ instead of $V_{\text{reset}} < 0$. The continuous transition from $f = 0$ to $f \neq 0$ becomes discontinous, as seen in figure 3. Such an f-I curve results in different computational properties than those displayed by class I neurons (as defined in figure 2). Whereas class I neurons encode inputs continuously with respect to their frequency, class 2 neurons jump their frequency discontinuously at the critical input $I = 0$. BOTH of the f-I curves for the single neuron have an undefined slope at the critical current $I = 0$.

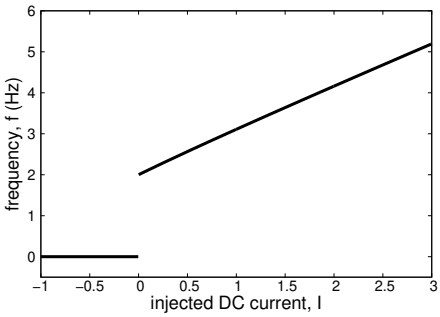

Figure 3: Equation 2 with $V_{\text{reset}} = 1$ and $V_{\text{peak}} = 2$, showing class II excitability in the standard QIF neuron ($V_{\text{reset}} > 0$).

Are there possible states a neuron injected with constant current may adopt, other than spiking with constant frequency? In fact, the standard QIF neuron's steady-state is determined entirely by the the input current I and the reset voltage V_{reset}. To see this, let us examine the equilibrium voltage values V^*, which are given by the fixed points of equation 1:

$$0 = I + (V^*)^2$$
$$V^* = \pm\sqrt{-I}$$

This implies that equilibrium voltages only exist when $I < 0$. Since they are equal in magnitude, let us refer to the positive equilibrium as V^* so that the negative equilibrium simply becomes $-V^*$. To investigate their stability, define the function $f(V) \equiv I + V^2$. Taking its derivative and plugging in the equilibria yields

$$f'(V) = 2V$$
$$f'(V^*) = 2\sqrt{-I}$$
$$f'(-V^*) = -2\sqrt{-I}$$

When they exist ($I < 0$), the positive equilibrium voltage is unstable and the negative equilibrium is stable. This is confirmed graphically via figures 4 and 5. The time series in figure 5 corresponding to the phase portrait in figure 4 demonstrates that a threshold voltage exists at V^* when $I < 0$. For initial conditions above this threshold (blue trace in figure 5), the neuron's voltage spikes to V_{peak} before returning to its resting state ($-V^*$). For an initial condition below this threshold (orange trace in figure 5), the neuron's voltage monotonically decreases to its resting state.

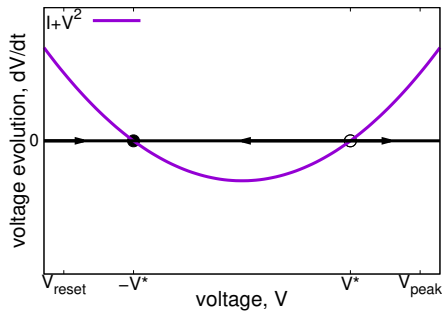

Figure 4: Phase portrait of QIF neuron. $I < 0$ and $V_{\text{reset}} < V^*$. Arrows indicate the local direction of the flow.

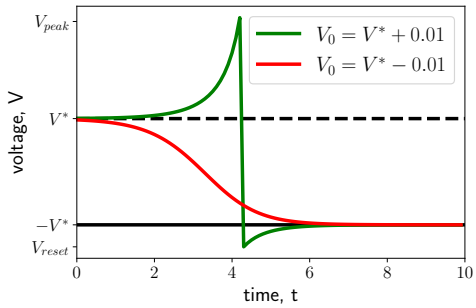

Figure 5: Time series of standard QIF neuron for two different initial conditions, corresponding to the parameter values in figure 4. $V^* = \sqrt{-I}$ plays the role of threshold here.

Suppose that we maintain the same input current I as in figures 4 and 5, hence keeping V^* fixed,

but tune the reset voltage so that $V_{\text{reset}} > V^*$ instead of $V_{\text{reset}} < V^*$. The QIF neuron's dynamics are then described by figures 6 and 7. The transient "up" state in figure 5 consisting of one spike before going to rest is replaced by a perpetual up state of spiking with constant frequency in figure 6. This is because the neuron's voltage is caught in an endless loop of increasing to V_{peak}, resetting to V_{reset}, and then back to increasing again. The down state of figure 4 is identical to that of figure 6, since the dynamics for $V < V^*$ do not change when tuning the reset voltage so that $V_{\text{reset}} > V^*$.

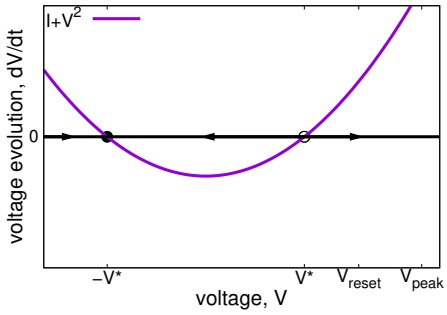

Figure 6: Phase portrait of the QIF neuron, given by equation 1. In this plot, $I < 0$ and $V_{\text{reset}} > V^*$. Once $V > V^*$, the voltage is caught in an endless cycle between V_{reset} and V_{peak}. This represents a "tonic" (constant frequency) spiking state.

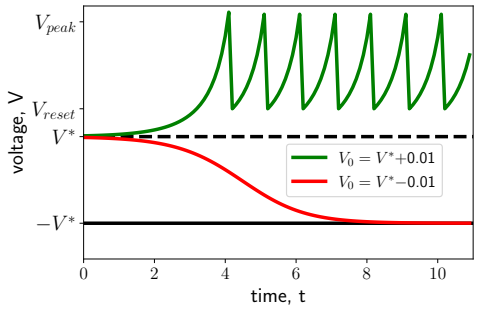

Figure 7: Time series of standard QIF neuron for two different initial conditions, corresponding to the parameter values in figure 6. $V^* = \sqrt{-I}$ plays the role of threshold here.

Self-coupled QIF neuron

To begin the process of extending the standard QIF model of a single neuron to a network, we must introduce the notion of synaptic coupling. Consider a QIF neuron that is coupled to itself in the following way [2]:

$$\dot{V} = I + V^2 + Js \tag{4a}$$

$$\dot{s} = -\frac{s}{\tau_s} \tag{4b}$$

where $s > 0$ is a synaptic variable providing positive feedback to the voltage V at a time scale $\tau_s > 0$. The parameter $J > 0$ controls the magnitude of the synaptic coupling. As before, I is the injected current. The reset rule is: if $V \geq V_{\text{peak}}$, then

$$V \to V_{\text{reset}},$$
$$s \to s + 1$$

Similarly to the standard QIF neuron, the self-coupled QIF neuron can display bistability for negative constant currents, as shown in figures 8 and 9. Furthermore, this model is capable of displaying spiking with a frequency that increases in time, even for a constant input. This happens through a build-up of the synaptic variable s when $\tau_s > 1$, as shown in figure 10.

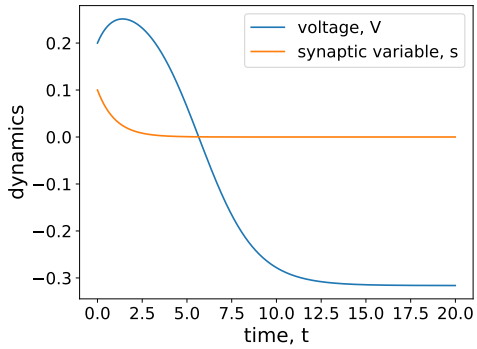

Figure 8: Time series of self-coupled QIF neuron (equation 4). Parameter values are $I = -0.1$, $V_{\text{reset}} = -1$, $V_{\text{peak}} = 2$, $J = 1.5$, and $\tau_s = 1$. Initial conditions are $V_0 = 0.2$

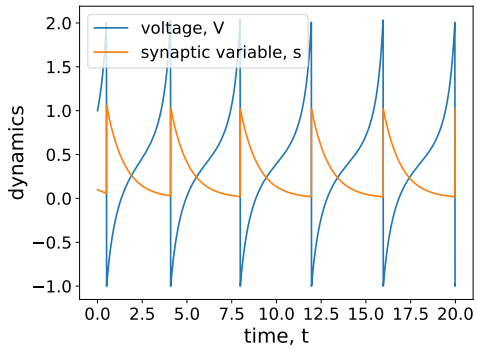

Figure 9: Time series of self-coupled QIF neuron (equation 4). Parameter values are identical to those of figure 7, except the initial voltage $V_0 = 1$.

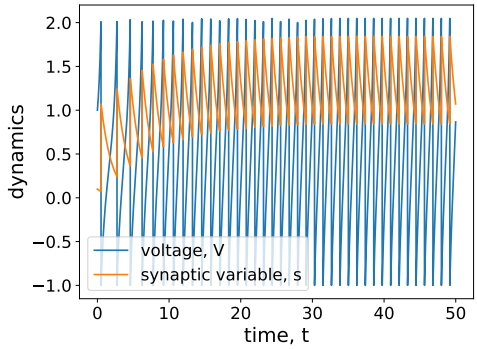

Figure 10: Time series of self-coupled QIF neuron (equation 4). Parameter values are $I = -0.1$, $V_{\text{reset}} = -1$, $V_{\text{peak}} = 2$, $J = 1.5$, and $\tau_s = 1.5$. Initial conditions are $V_0 = 1$, $s_0 = 0.1$.

The f-I curves for the self-coupled QIF neuron are shown below in figure 11. These were obtained by solving equation 4 numerically for 200 total time steps, counting the number of spikes in the last 100 time steps, and dividing by 100. This procedure was carried out for each of the injected DC currents shown. Note the similarity of the curves shown in figure 11 with those of figure 2. Increasing J in the self-coupled QIF neuron model increases the slope of the f-I curve (results not shown).

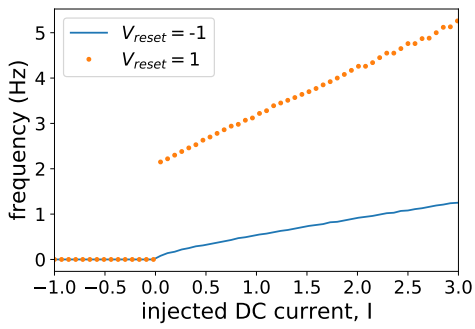

Figure 11: Asymptotic frequency versus injected DC current for the self-coupled QIF neuron with $J = 0.1$, $\tau_s = 1.5$, and $V_{\text{peak}} = 2$.

Network of excitatory QIF neurons

Following Avitabile *et al.* [1], the self-coupled QIF neuron represented by equation 4 is now generalized to a globally-coupled network of N excitatory QIF neurons:

$$\dot{V}_j = V_j^2 + I_j \tag{5a}$$

$$\dot{s}_j = -\frac{s_j}{\tau_s} \tag{5b}$$

where $j = 1, \ldots, N$. The reset rule is

$$\text{if } V_j \geq V_{\text{peak}}, \text{ then } \begin{cases} V_j \to V_{\text{reset}} \\ s_j \to s_j + \frac{1}{N} \end{cases}$$

where $I_j = \eta_j + Js + I(t)$ is the input to the jth neuron. The quantity η_j is a random variable representing the heterogeneous component of the input that is unique to the jth neuron. The function $I(t)$ is a time-dependent input to all neurons. The synaptic coupling J multiplies the average synaptic activation $s = s(t)$ given by

$$s(t) = \frac{1}{N} \sum_{j=1}^{N} \sum_{k} \int_{-\infty}^{t} \frac{e^{-(t-t')/\tau_s}}{\tau_s} \delta(t' - t_j^k) \mathrm{d}t'$$

where t_j^k is the kth spike of the jth neuron and $\delta(t)$ is the Dirac delta function with the property

$$h(t) = \int_{-\infty}^{\infty} h(t')\delta(t - t')\mathrm{d}t'$$

for any function h.

Montbrio *et al.* [11] take the thermodynamic limit $N \to \infty$ of equation 5, which converts the discrete neurons with all of their voltages into the conditional probability density function $\rho(V|\eta, t)$. The quantity $\rho(V|\eta, t)\mathrm{d}V$ is interpreted as the fraction of neurons with voltage between V and $V + \mathrm{d}V$, for input η at time t. Since the number of neurons is constant in this model, the evolution of this quantity is given by a continuity equation. Under the assumption that solutions to this continuity equation converge to a Lorenztian distribution, referred to as the "Lorentzian ansatz (LA)" in ref. [11], the authors derive the mean-field equations

$$\dot{r} = \frac{\Delta}{\pi} + 2rv \tag{6a}$$

$$\dot{v} = v^2 + \bar{\eta} + Jr + I(t) - \pi^2 r^2 \tag{6b}$$

where r is the average firing rate (frequency) of the network, and v is its average voltage. The distribution of the heterogeneous input η has half-width Δ and mean $\bar{\eta}$. The transcendental number π is the familiar 3.14..., not a parameter. In lieu of the spike-resetting of the QIF model, the term $-\pi^2 r^2$ restricts v from blowing up to ∞. Equation 6 is studied via mathematical analysis and numerical simulation in the results section below.

Mean-field model of excitation/inhibition

In a fashion very similar to that outlined in the previous subsection, Montbrio *et al.* [11] derive a mean-field model for the dynamics of excitatory AND inhibitory neuronal populations from a QIF framework, eventually arriving at the equations

$$\dot{r}_e = \frac{\Delta_e}{\pi} + 2r_e v_e \tag{7a}$$

$$\dot{v}_e = v_e^2 + \bar{\eta}_e + J_{ee} r_e - J_{ie} r_i + I_e(t) - (\pi r_e)^2 \tag{7b}$$

$$\dot{r}_i = \frac{\Delta_i}{\pi} + 2r_i v_i \tag{7c}$$

$$\dot{v}_i = v_i^2 + \bar{\eta}_i + J_{ei} r_e - J_{ii} r_i + I_i(t) - (\pi r_i)^2 \tag{7d}$$

where the subscript i stands for inhibitory and e stands for excitatory. All symbols have the same interpretation as in the model of equation 6. The synaptic coupling strengths J_{xy} where $x, y \in \{e, i\}$ are the main focus here. These equations are studied numerically in the results section below.

Results

Excitatory mean-field model

The divergence of the vector field given by equation 6 is

$$\frac{\partial \dot{r}}{\partial r} + \frac{\partial \dot{v}}{\partial v} = 4v$$

which is non-zero for all r and all $v \neq 0$. By Bendixson's criterion [10], this means that there are no closed orbits for $v > 0$, nor for $v < 0$, since $\nabla \cdot \dot{\boldsymbol{x}} = 4v$ doesn't change sign on these regions. The excitatory mean-field model therefore has no limit-cycle attractors in the absence of time-varying external inputs. The nullclines $\dot{r} = 0$ and $\dot{v} = 0$ are given by:

$$v = -\frac{\Delta}{2\pi r} \qquad (\dot{r} = 0)$$

$$v = \pm\sqrt{\pi^2 r^2 - Jr - \bar{\eta} - I} \qquad (\dot{v} = 0)$$

which are plotted in figures 12-16, along with a few sample trajectories in the phase plane. The equilibria (r_∞, v_∞) where both $\dot{r} = 0$ and $\dot{v} = 0$ are given by the intersections of the nullclines. They are the solutions to the algebraic system stated directly above. Negative frequencies $r < 0$ are disregarded,

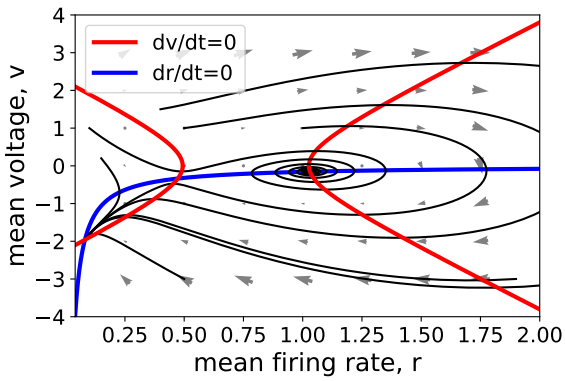

Figure 12: Nullclines of model 6 shown with the thick colored curves, 5 sample trajectories shown with thin black curves. The parameter values are $I = 0$, $\Delta = 1$, $\bar{\eta} = -5$, and $J = 15$. Vector field shown in grey.

since they are unphysical. Three physically-viable fixed points are visualized in figure 12. For the choice of parameter values shown, there appears to be a stable node around $(r_\infty, v_\infty) = (0.1, -2)$, a stable spiral around $(r_\infty, v_\infty) = (1, -0.2)$, and a saddle separating their basins of attraction around $(r_\infty, v_\infty) = (0.5, -0.2)$. Biologically, the saddle and its corresponding manifolds are an effective threshold that separates low-frequency, negative-voltage networks from high-frequency, zero-voltage networks.

Mathematically, the effect of increasing the injected DC current from 0 is to shift the left branch of the v-nullcline (the red curve on the left of figure 12) to the left. If one were to keep doing this, the left branch of the red curve would eventually intersect the blue curve tangentially at exactly one point. This moment of tangent intersection is a saddle-node bifurcation, and numerical evidence for its existence in provided in figures 13 and 14 below. The biological implication of this is that raising the injected DC current high enough causes the network to tend to a high-frequency steady state, regardless of the initial condition. The average voltage of this steady state is slightly less than zero, meaning that the actual voltages of all the neurons are fluctuating evenly around $v \approx 0$.

Although the saddle-node bifurcation shown in figure 13 has been induced by modifying the input current I, altering the mean of the heterogeneous input $\bar{\eta}$ can also cause this bifurcation to occur. This is evident by observing that varying I in the $\dot{v} = 0$ equation shifts the nullcline in the same way that altering $\bar{\eta}$ by the same amount would.

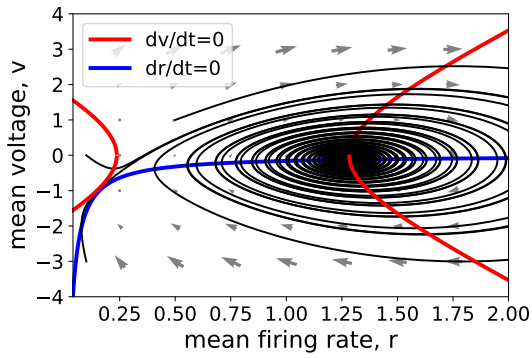

Figure 13: Nullclines and sample trajectories for model 6, with the same parameter values as those of figure 12, except $I = 2$. The increased input I has induced a saddle-node bifurcation, indicated by the tangent intersection of the nullclines on the left of the graph. Vector field shown in grey.

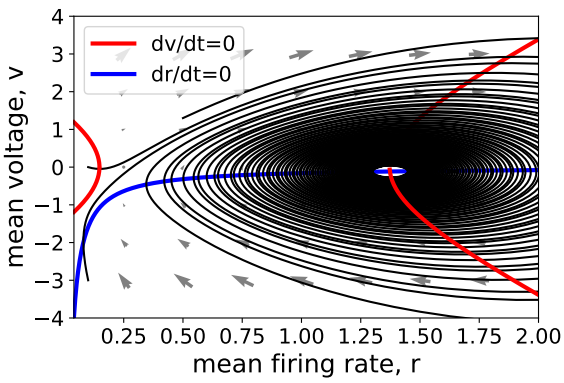

Figure 14: Nullclines and sample trajectories for model 6, with the same parameter values as those of figure 12, except $I = 3$. Increasing the input I past its bifurcation value has caused a complete loss of the low-frequency asmyptotic state, leaving only the high-frequency steady state with v slightly less than 0.

The case $I = 0$ is now considered and an analytical form for the parameters $(J_{\text{SN}}, \bar{\eta}_{\text{SN}})$ that yield a saddle-node bifurcation in this case is sought. Also assume fixed $\Delta = 1$. Since a saddle-node bifurcation occurs when the nullclines intersect tangentially, and the bifurcation occurs on the negative branch of the v nullcline in figure 13, we require

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(-\frac{\Delta r^{-1}}{2\pi}\right) = \frac{\mathrm{d}}{\mathrm{d}r}\left[-(\pi^2 r^2 - J_{\text{SN}} r - \bar{\eta}_{\text{SN}})^{1/2}\right]$$

$$\frac{\Delta}{2\pi r^2} = \frac{J_{\text{SN}} - 2\pi^2 r}{2\sqrt{\pi^2 r^2 - J_{\text{SN}} r - \bar{\eta}_{\text{SN}}}}$$

But at the intersection of the nullclines shown on the left of figure 13, we have that $v = -\Delta/(2\pi r)$ and $v = -\sqrt{\pi^2 r^2 - Jr - \bar{\eta}}$. Substituting the former into the left-hand side above and the latter into the right-hand side yields

$$-\frac{v}{r} = \frac{2\pi^2 r - J_{\text{SN}}}{2v}$$

Substituting $v^2 = \Delta^2/(2\pi r)^2$ from the r nullcline and solving for J_{SN} yields

$$J_{\text{SN}} = 2\pi^2 r + \frac{\Delta^2}{2\pi^2 r^3}$$

Plugging this into the equation representing an intersection of the nullclines

$$-\frac{\Delta}{2\pi r} = -\sqrt{\pi^2 r^2 - J_{\text{SN}} r - \bar{\eta}_{\text{SN}}}$$

yields

$$\frac{\Delta}{2\pi r} = \sqrt{\pi^2 r^2 - \left(2\pi^2 r + \frac{\Delta^2}{2\pi^2 r^3}\right) r - \bar{\eta}_{\text{SN}}}$$

$$\vdots$$

$$\bar{\eta}_{\text{SN}} = -\pi^2 r^2 - \frac{3\Delta^2}{(2\pi r)^2} \qquad \blacksquare$$

Which concludes the demonstration of where the black curve in the bifurcation diagram of figure 17 comes from.

There is also a transition directly from the globally-stable spiral of figure 14 to a globally-stable node. Numerical evidence for this claim is shown in the series of figures 15 and 16.

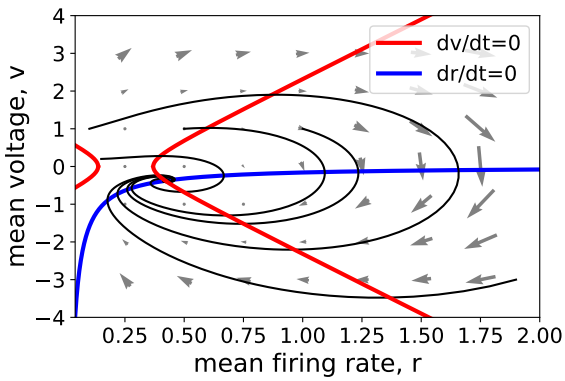

Figure 15: Parameter values are $I = 0$, $\Delta = 1$, $\bar{\eta} = -0.5$, and $J = 5$. Colors as before. The spiral is globally stable.

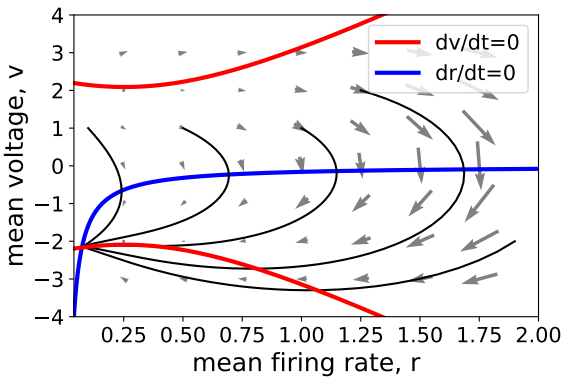

Figure 16: Parameter values are $I = 0$, $\Delta = 1$, $\bar{\eta} = -5$, and $J = 5$. Colors as before. The node is globally stable.

Where exactly does the transition from globally-stable spiral to globally-stable node occur, in parameter space? The answer to this question lies in the eigenvalues of the fixed points, which are found by considering the stability matrix of partial derivatives, call it S:

$$S \equiv \begin{pmatrix} \frac{\partial \dot{r}}{\partial r} & \frac{\partial \dot{r}}{\partial v} \\ \frac{\partial \dot{v}}{\partial r} & \frac{\partial \dot{r}}{\partial v} \end{pmatrix} = \begin{pmatrix} 2v & 2r \\ J - 2\pi^2 r & 2v \end{pmatrix}$$

which has trace $\mathrm{tr}(S) = 4v$ and determinant $\det(S) = 4v^2 + (2\pi^2 r - J)(2r)$. The eigenvalues λ are given by the characteristic equation

$$\lambda^2 - \mathrm{tr}(S)\lambda + \det(S) = 0$$
$$\lambda = \frac{\mathrm{tr}(S) \pm \sqrt{\mathrm{tr}(S)^2 - 4\det(S)}}{2}$$

Since the r nullcline is defined only for $v < 0$, $\mathrm{tr}(S) = 4v < 0$ for all equilibria. The transition between nodes and spirals occurs when the discriminant of the characteristic equation (the quantity under the square root, above) equals 0. Symbolically,

$$\mathrm{tr}(S)^2 - 4\det(S) = 0$$
$$(4v)^2 - 4(4v^2 + (2\pi^2 r - J)(2r)) = 0$$
$$\vdots$$
$$J = 2\pi^2 r$$

But at the fixed point shown in figures 15 and 16, the nullclines intersect:

$$-\frac{\Delta}{2\pi r} = -\sqrt{\pi^2 r^2 - Jr - \bar{\eta}}$$

Squaring, substituting $r = J/(2\pi^2)$ from above, and relabeling $\bar{\eta} \to \bar{\eta}_f$ yields

$$\left(\frac{\Delta\pi}{J}\right)^2 = \left(\frac{J}{2\pi}\right)^2 - \frac{J^2}{2\pi^2} - \bar{\eta}_f$$
$$\vdots$$
$$\bar{\eta}_f = -\left(\frac{J}{2\pi}\right)^2 - \left(\frac{\Delta\pi}{J}\right)^2$$

which is the boundary in parameter space that separates stable spirals from stable nodes. A bifurcation diagram summarizing the above analysis is shown below in figure 17.

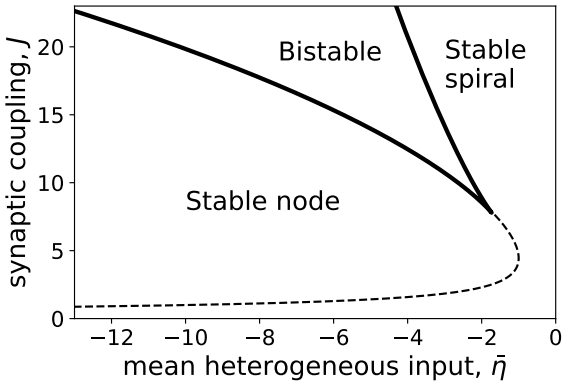

Figure 17: Bifurcation diagram for model 6, with $\Delta = 1$. The thick solid curve containing the bistable regime is given parametrically by $(\bar{\eta}_{\mathrm{SN}}, J_{\mathrm{SN}}) = \left[-\pi^2 r^2 - 3\Delta^2/(2\pi r)^2, 2\pi^2 r + \Delta^2/(2\pi^2 r^3)\right]$. The dashed curve separating nodes from spirals is given by $\bar{\eta}_f = -[J/(2\pi)]^2 - (\Delta\pi/J)^2$.

A numerically-derived f-I curve for this model is shown in figure 18. Although the mean-field network is composed of neurons with f-I curves containing cusps and discontinuities (figure 11), the f-I curve of the network is both smooth and continuous.

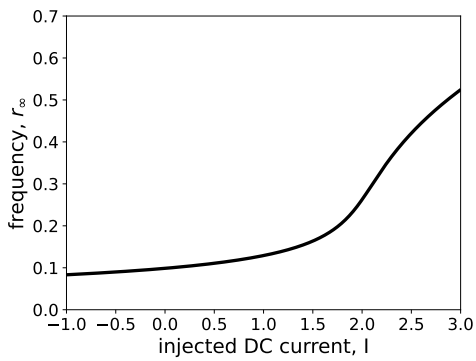

Figure 18: The f-I curve for model 6, with $\Delta = 1$, $\bar{\eta} = -3$, $J = 5$. In contrast to the f-I curves of figures 2 and 11, this one is both smooth AND continuous on the range of I considered.

Excitatory/inhibitory mean-field model

The model of equation 7 is now briefly investigated through numerical simulations. The focus in this subsection is on demonstrating how the asymptotic frequencies (firing rates) of the excitatory and inhibitory populations depend on the "balance" of the network. The relative synaptic coupling strengths (the J's) are interpreted as quantities that reflect this network balance.

The dynamics of a perfectly-balanced network ($J_{ei} = J_{ie}$ and $J_{ii} = J_{ee}$) are shown in figure 19. Unsurprisingly, the macroscopic observables r and v quickly converge to the same steady state for both populations in the absence of external inputs ($I_e(t) = I_i(t) = 0$).

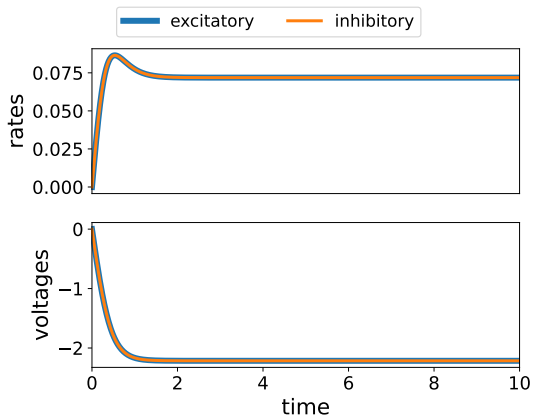

Figure 19: Time series of model 7 with parameter values $\Delta_e = \Delta_i = 1$, $\bar{\eta}_e = \bar{\eta}_i = -5$, $J_{ie} = J_{ei} = J_{ee} = J_{ii} = 2$, and $I_e(t) = I_i(t) = 0$.

The excitability of the network becomes non-trivial when it is not perfectly balanced. Suppose that network balance is broken in the following way. Let us assume equal interactions between populations through a mutual coupling $J_{ie} = J_{ei} \equiv K$. The unbalancing comes from an asymmetry in self-interactions $J_{ee} \neq J_{ii}$. Let us specifically impose $8J_{ii} = 2J_{ee}$ to reflect the fact that around 20% of the neurons in cortex are inhibitory [13]. An analog to the f-I curves from earlier is shown in figure 20 for such a scenario. The asymptotic frequency of the network is shown for each population, across a range of coupling strengths K. This result shows that the excitatory population always has a higher asymptotic frequency for low and negative coupling. For coupling $K > 1.7$ or so, however, there are coupling strengths that favor the inhibitory population. This goes on for higher values of K in an alternating fashion.

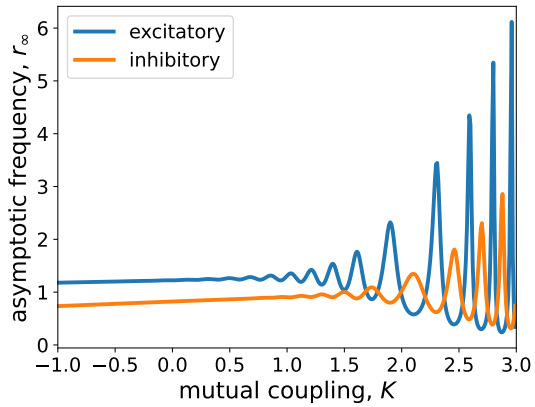

Figure 20: Excitability curves for an unbalanced network $J_{ii} = 2$, $J_{ee} = 8$ with mutual coupling $J_{ei} = J_{ie} \equiv K$ (see text). Other parameter values are $\Delta_e = \Delta_i = 1$, $\bar{\eta}_e = \bar{\eta}_i = 5$, and $I_e(t) = I_i(t) = 0$.

Discussion

It has been shown that similar patterns of excitability persist across neuronal systems of varying spatial scales, from the single-cell level to that of globally coupled, heterogeneous populations. The introduction of inhibition produces a rather complicated pattern of excitability, as demonstrated in figure 20. Upon first inspection, this curve looks fundamentally different than those shown for the single neuron/population cases depicted in figures 2, 11, and 18. If the criterion by which we judge their distinction is monotonicity, this is quite true. On the other hand, all of these curves have the property that low stimulation/coupling results in low activity that varies insignificantly as the stimulation is slowly increased. Then, once the stimulation is high enough, there is an abrupt change in the response of the system from low asymptotic activity to high. This result forms the pinnacle of the present work and points to a possible characteristic of large populations of neurons with both excitation and inhibition. Namely, if the mutual coupling K were strong enough (on the right half of figure 20) and even slightly oscillatory in nature (perhaps due to outside influences, such as neuromodulators), the average frequencies of the excitatory and inhibitory populations would alternate significantly in time. Such a regime constitutes a "bursting" network. On the other hand, if the coupling K is too weak (on the left side of figure 20), then the populations have roughly constant frequencies and correspond to a "tonic" network. This manifestation of tonic versus

bursting at the large network scale contrasts with that of the single-cell and small network level [4], which inherently relies on ion channel activation operating on a separation of time scales. Here, the tonic versus bursting modes are instead due to an asymmetry in the network that is amplified by a mutual coupling.

Theoretical and computational studies of neuronal population activities play a vital role in understanding the complex measurements of nervous systems made by devices like multi-electrode arrays (MEA), electroencephalograms (EEG), and functional magnetic resonance imaging (fMRI) in both scientific and clinical settings. These technologies capture spatiotemporally-averaged data that is comprehended optimally when analyzed in conjunction with a tractable quantitative framework. This practice is particularly valuable for researching the collective properties of neurons. Such properties are pertinent to complex functions of the nervous system like vision, where the effective computational unit is not the single neuron, but rather columns and layers of cortex that correspond to the recognition of image features of various complexity. Such a paradigm is an example of a possible extension of this model, especially since it accounts for excitation AND inhibition: an interplay that is intellectually inescapable when conceptualizing the process of visual perception [12]. The analysis of constant inputs investigated in this manuscript also forms a groundwork for the potential extension to time-varying inputs, to which some preliminary efforts have been devoted in the supplementary information. Finally, an analysis of the extent to which "global coupling" is a safe assumption under various experimental contexts is warranted.

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